

From Image Enhancement to Automated Galaxy Morphology

An interpretable pipeline for structure and apparent spiral handedness in JWST deep-field imagery

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Abstract

Can weak galaxy structure be recovered from small, noisy image crops well enough to classify morphology and infer an apparent spiral winding direction? This project began as a sequence of image-enhancement experiments: Gaussian background subtraction, unsharp masking, binary line extraction, bilateral denoising, logarithmic stretching and Sobel gradients. Those experiments made faint features easier to see, but also showed that visual enhancement can amplify pixel boundaries, noise and unrelated structure. The project therefore moved from “making the image look clearer” to measuring reproducible quantities. The final pipeline estimates the sky background, isolates a connected source, measures size, axis ratio, concentration, asymmetry and clumpiness, assigns interpretable morphology scores, and independently tests spiral structure with a log-polar Fourier method using $m = 2, 3, 4$ modes. A robustness layer perturbs the center, analysis annulus, smoothing scale and image noise, and performs a mirror-consistency test before accepting clockwise (CW) or counter-clockwise (CCW) apparent winding. A batch extension detects sources in a full JWST field, maps detections back to native coordinates, saves lossless crops, and builds a catalog automatically. In a 120-candidate pilot run, 118 objects entered the morphology statistics: 87 were uncertain, 11 edge-on disks, 9 spirals, 6 irregulars, 4 merger/disturbed candidates and 1 elliptical. None passed every strict CW/CCW criterion. The result is therefore a conservative, interpretable baseline rather than a trained classifier or a claim of cosmic handedness asymmetry.

1 Introduction and scope

Deep astronomical images contain galaxies spanning a wide range of angular sizes, orientations, surface brightnesses and signal-to-noise ratios. For a visually clear nearby spiral, a human can often identify arms and their apparent winding direction immediately. A much harder case is a distant object represented by only a few tens of useful pixels. The initial question of this project was deliberately practical: can image processing reveal enough structure in such a crop to determine whether a galaxy is spiral, and if so, whether its visible arms wind clockwise or counter-clockwise on the image?

The project did not begin with a pre-trained neural network. Instead, it developed from simple image-processing operations into an interpretable measurement pipeline. This progression was useful because each failure mode suggested the next algorithmic step. Gaussian residuals made local structure visible but also enhanced noise. Binary thresholding produced clean-looking lines but discarded photometric information. Bilateral filtering preserved edges better than Gaussian smoothing, yet Sobel gradients often emphasized the pixel grid rather than genuine arms. These observations motivated a shift toward source detection, robust morphological measurements and signed Fourier evidence.

The final workflow has four goals. First, it should reject cases in which the source is too small or poorly isolated. Second, it should describe morphology with interpretable quantities rather than only a label. Third, a spiral-handedness claim should survive moderate changes in analysis choices. Fourth, the same logic should scale from one manually cropped galaxy to a full field containing many objects.

The word *handedness* is used here in an image-plane sense only: CW and CCW refer to the direction in which a visible arm turns while moving outward from the estimated center. This is not, by itself, a measurement of the galaxy’s three-dimensional physical rotation. Human classifications of spiral handedness are also known to be vulnerable to bias, so conservative acceptance and mirror tests are important when searching for population-level asymmetries [4].

The present study is a development and pilot-validation paper. The input is a processed RGB JWST composite rather than calibrated multi-filter science data. NASA notes that public full-color Webb images are stretched, aligned and combined for visualization [5]; the next scientific stage should therefore repeat the analysis on calibrated FITS products from the JWST pipeline [6]. No claim is made here that the current heuristic class scores are probabilities.

2 From visual enhancement to measurable structure

2.1 First experiment: Gaussian residuals and line extraction

The first code attempted to reveal faint structure by separating a smooth component from high-frequency detail. Each RGB channel was convolved with a Gaussian kernel of width $\sigma = 7$ pixels. The residual was

$$R_\sigma(x, y) = I(x, y) - (G_\sigma * I)(x, y), \quad (1)$$

then multiplied by a gain factor of three and clipped to the display range. A brightness mask retained residual pixels above a threshold. A second local smoothing step produced an unsharp correction,

$$U = R_\sigma + \alpha [R_\sigma - G_{\sigma_u} * R_\sigma], \quad (2)$$

with $\alpha \simeq 2$ and $\sigma_u \simeq 2$ pixels. Finally, thresholding and additional smoothing converted the residual into a line-like representation.

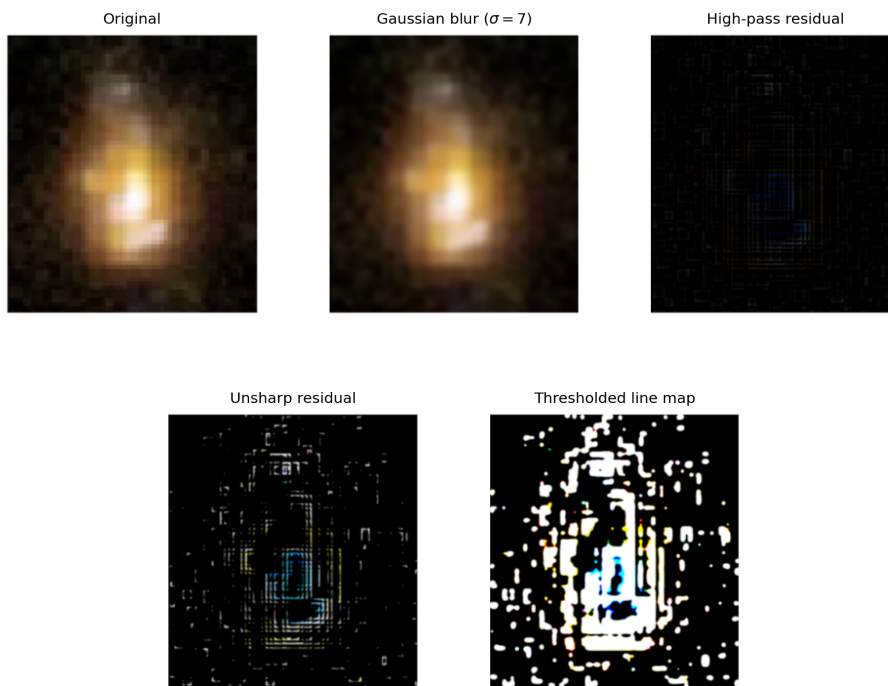


Figure 1: First enhancement experiment, reformatted as a diagnostic sequence rather than a notebook screenshot. The upper row shows the original crop, Gaussian smoothing, and the high-pass residual; the lower row shows the unsharp residual and the thresholded line map. The last two panels illustrate why this stage was retained as a development experiment rather than as the final classifier: hard enhancement makes weak pixel-scale structure look deceptively definite.

Figure 1 was useful precisely because it exposed the weakness of this approach. The output became visually dramatic, but the final binary map no longer preserved how strong, coherent

or statistically significant a feature was. A square pixel grid and noise spikes could appear as “structure” after sharpening and thresholding. The procedure was therefore suitable as a visualization experiment, not as a final classifier.

Development lesson. High-pass filtering can reveal local detail, but a clearer-looking image is not necessarily a more informative measurement. Hard thresholds erase uncertainty and can create false confidence.

2.2 Second experiment: bilateral denoising, log stretch and gradients

The next version replaced part of the Gaussian-only workflow with edge-preserving bilateral denoising, following the idea of weighting nearby pixels by both spatial and intensity similarity [1]. Each channel was denoised with a bilateral filter, sharpened with an unsharp mask, and then transformed with a logarithmic stretch,

$$I_{\log} = \frac{\ln(1 + \kappa I)}{\ln(1 + \kappa)}, \quad (3)$$

where the code used $\kappa = 9$ after normalizing each channel. Sobel derivatives estimated horizontal and vertical intensity changes, and the displayed gradient magnitude was

$$|\nabla I| = \sqrt{(S_x * I)^2 + (S_y * I)^2}. \quad (4)$$

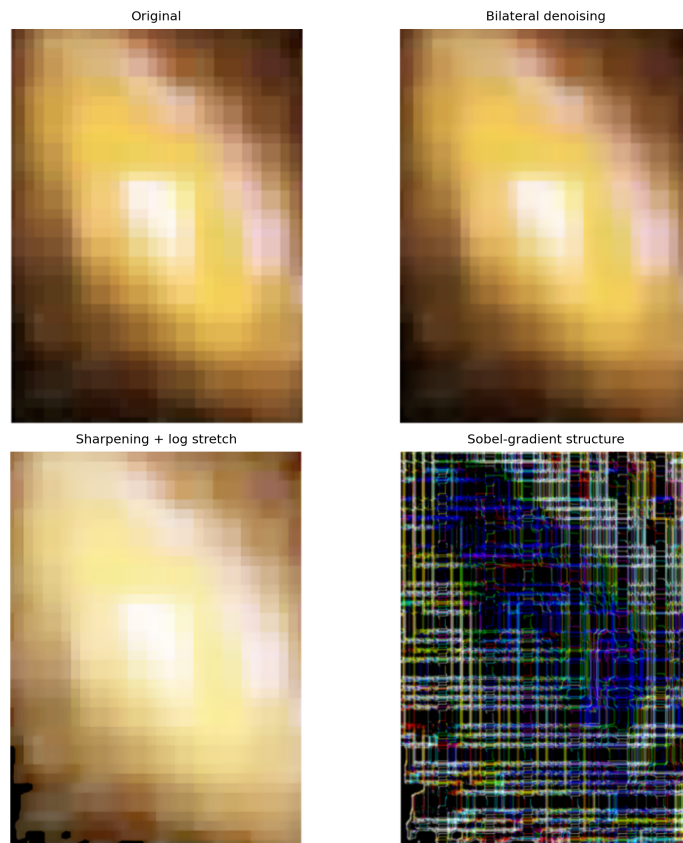


Figure 2: Second enhancement experiment. Bilateral denoising preserves the broad source profile while the log stretch increases visible contrast. The Sobel-gradient panel, however, is dominated by the sampling grid in this small crop. This failure motivated the transition from visual edge enhancement to quantitative morphology and log-polar Fourier measurements.

The log-stretched image often looked more interpretable to the eye than the first binary representation, but the gradient panel in Figure 2 revealed another problem: at low angular resolution, edges of enlarged detector pixels and compression blocks can dominate the derivative. This is a general

warning against treating image enhancement as information recovery. No filter can reconstruct arm geometry that is absent from the sampled data.

Development lesson. Denoising and contrast stretching are useful preprocessing tools, but classification needs measurements that remain meaningful when visualization parameters change. This motivated a switch from edge images to source geometry and signed residuals.

3 Quantitative morphology

3.1 Background, source mask and center

The final morphology stage first converts the RGB crop to luminance,

$$Y = 0.2126R + 0.7152G + 0.0722B, \quad (5)$$

and applies a robust linear scaling between the 1st and 99.5th percentiles. The sky level is estimated from a border region. Instead of a standard deviation, the code uses a median absolute deviation (MAD) estimator,

$$\sigma_{\text{sky}} = 1.4826 \text{ median} (|Y - \text{median}(Y)|), \quad (6)$$

which is less sensitive to isolated bright pixels.

A mildly smoothed image defines candidate source pixels above roughly $B + 3\sigma_{\text{sky}}$, where B is the estimated background. Connected-component labeling then selects the component associated with the central bright source. The center is not assumed to be the center of the crop: a weighted centroid is calculated around the central peak. The code records the number of source pixels, a peak signal-to-noise proxy and an integrated proxy

$$\text{SNR}_{\text{proxy}} = \frac{F_{\text{source}}}{\sigma_{\text{sky}} \sqrt{N_{\text{pix}}}}. \quad (7)$$

Cumulative flux radii $r_{20}, r_{50}, r_{80}, r_{90}$ are then measured. In this project, r_{90} became particularly useful as a practical resolution flag: an object may have high SNR while still being too small to contain enough independent spatial information for detailed morphology.

3.2 Shape, concentration, asymmetry and clumpiness

The source's second central moments form a 2×2 covariance matrix. If its eigenvalues are $\lambda_{\min}$ and $\lambda_{\max}$, the rough projected axis ratio is

$$q \equiv \frac{b}{a} = \sqrt{\frac{\lambda_{\min}}{\lambda_{\max}}}. \quad (8)$$

Values near unity describe round projected light distributions; small values indicate elongation and can signal an edge-on disk.

The concentration proxy is based on enclosed-light radii,

$$C = 5 \log_{10} \left(\frac{r_{80}}{r_{20}} \right). \quad (9)$$

The asymmetry proxy compares the image with a 180-degree rotation about the measured fractional-pixel center,

$$A \simeq \frac{\sum |I - I_{180}| - \text{sky correction}}{2 \sum I}. \quad (10)$$

Clumpiness is estimated from positive high-frequency residuals after subtracting a broader Gaussian-smoothed version of the source. These quantities are inspired by the concentration–asymmetry–clumpiness (CAS) framework used in quantitative galaxy morphology [2], but the implementation here is intentionally described as a *proxy*: its aperture choices, background corrections and thresholds have not yet been calibrated to reproduce a standard CAS catalog.

The pipeline also counts local bright components inside the connected source. This is not interpreted literally as a number of galactic nuclei; it is only an additional disturbance indicator.

3.3 Interpretable five-class morphology scores

The code maps measured features into bounded evidence variables such as roundness, elongation, smoothness, symmetry, disturbance and clumpiness. It then forms transparent class scores rather than using a black-box classifier. In compact notation, the current weights are

$$S_{\text{edge}} = 0.80E + 0.20Y, \quad (11)$$

$$S_{\text{ell}} = 0.25R + 0.30M + 0.20Y + 0.25C_h, \quad (12)$$

$$S_{\text{irr}} = 0.55D + 0.45K, \quad (13)$$

$$S_{\text{merge}} = 0.50D + 0.30P + 0.20K, \quad (14)$$

where E, R, M, Y, C_h, D, K, P denote normalized elongation, roundness, smoothness, symmetry, high concentration, disturbance, clumpiness and multiple-peak evidence. The spiral score additionally uses the Fourier evidence described in Sec. 4:

$$S_{\text{spiral}} = 0.20R + 0.45F_{\text{spiral}} + 0.10K_{\text{mid}} + 0.10Y + 0.15F_{\text{sign}}. \quad (15)$$

A class is not automatically accepted just because it has the largest score. In the current settings, the top score must exceed 0.42 and exceed the second-best score by at least 0.10; otherwise the final morphology is **UNCERTAIN**. Objects too small for reliable measurement become **UNRESOLVED**, while objects not reliably detected become **NO CLEAR SOURCE**.

Development lesson. Morphology became more useful when the program returned both a label and the measurements that produced it. An uncertain output is an intended scientific result, not a program failure.

4 Spiral structure and apparent handedness

4.1 Why log-polar coordinates?

A logarithmic spiral can be written

$$r = r_0 e^{b\theta}. \quad (16)$$

Taking a logarithm gives

$$\ln r = \ln r_0 + b\theta, \quad (17)$$

so a curved arm in Cartesian coordinates becomes approximately a straight ridge in $(\theta, \ln r)$ space. This is the central reason for moving from the early edge maps to a log-polar representation. Two-dimensional Fourier methods have long been used to quantify logarithmic spiral structure and pitch angles [3].

For each radius, the code subtracts the azimuthal mean. This removes much of the axisymmetric bulge/disk light while preserving signed residuals. An FFT along the angular direction provides Fourier components for arm modes $m = 2, 3, 4$. For a candidate logarithmic slope $k = d\theta/d \ln r$, the radial phase is matched through

$$A_m(k) = \sum_j w_j F_m(u_j) \exp[i m k (u_j - u_0)], \quad u_j = \ln r_j, \quad (18)$$

with a Hann radial weight w_j . The response power is $P_m(k) = |A_m(k)|^2$.

The strongest positive-slope response is labeled P_{CW} and the strongest negative-slope response P_{CCW} in the image-coordinate convention used by the code. Their signed margin is

$$H = \frac{P_{\text{CW}} - P_{\text{CCW}}}{P_{\text{CW}} + P_{\text{CCW}}}. \quad (19)$$

Thus $H > 0$ favors CW and $H < 0$ favors CCW, while $H \approx 0$ means that the image does not clearly distinguish the two.

4.2 Rejecting false spirals

A single maximum Fourier response is not enough. A bar, diffraction feature or elongated light distribution can imitate part of a spiral template. The pipeline therefore records the ratio of the best nonzero-slope template to a straight ($k = 0$) template, the fraction of residual structure explained by the best mode, radial phase coherence and arm amplitude relative to the measured sky noise.

The code then repeats the measurement for five nearby centers, two smoothing scales and three annuli, giving 30 parameter trials. The median signed score is used, and the *parameter stability* is the fraction of trials that preserve a meaningful sign. The pilot batch run also adds eight Gaussian-noise perturbations to each robust candidate. Finally, the image is mirrored horizontally. A correct implementation should reverse the signed handedness, so the code checks

$$\epsilon_{\text{mirror}} = |H(I) + H(I_{\text{mirror}})|. \quad (20)$$

Table 1 lists the present strict thresholds. These numbers are engineering choices for the pilot, not values established from an external labeled validation set.

Table 1: Strict apparent-handedness criteria in the current pipeline. Every listed diagnostic can veto a CW/CCW acceptance.

Diagnostic	Acceptance requirement
Signed margin	$ H \geq 0.35$
Parameter stability	≥ 0.90
Noise stability	≥ 0.90
Spiral phase coherence	≥ 0.45
Explained residual structure	≥ 0.20
Spiral/straight response	≥ 1.30
Arm amplitude / sky noise	≥ 0.45
Projected axis ratio	$b/a \geq 0.45$
Mirror consistency	$\epsilon_{\text{mirror}} \leq 0.03$

Development lesson. The goal changed from “find the strongest spiral-looking feature” to “accept a direction only if the sign survives reasonable perturbations and alternative explanations.”

5 Automatic field detection and native-resolution cropping

The single-object analyzer was then extended to an entire deep field. A computational copy of the field may be reduced to a maximum side of 2600 pixels for source detection, but this resized image is never used as the saved science crop. After estimating a broad background with a Gaussian scale proportional to 2.5% of the smaller image dimension, the detector thresholds a high-pass residual at approximately

$$T = \text{median}(R) + 2.8 \sigma_{\text{MAD}}(R). \quad (21)$$

Binary closing and dilation join nearby significant pixels. Connected components smaller than 10 detection pixels or narrower than three pixels are rejected. Each surviving component is assigned a square crop whose side is roughly 3.2 times the detected extent, clipped between 48 and 500 native pixels.

The crucial detail is coordinate mapping. Detection coordinates are transformed back to the original image, and `PIL.Image.crop` extracts the object directly from the native image. The crop is saved as PNG without resizing or lossy re-compression. This preserves the original pixel information inside the crop, while keeping enough surrounding sky for background estimation.

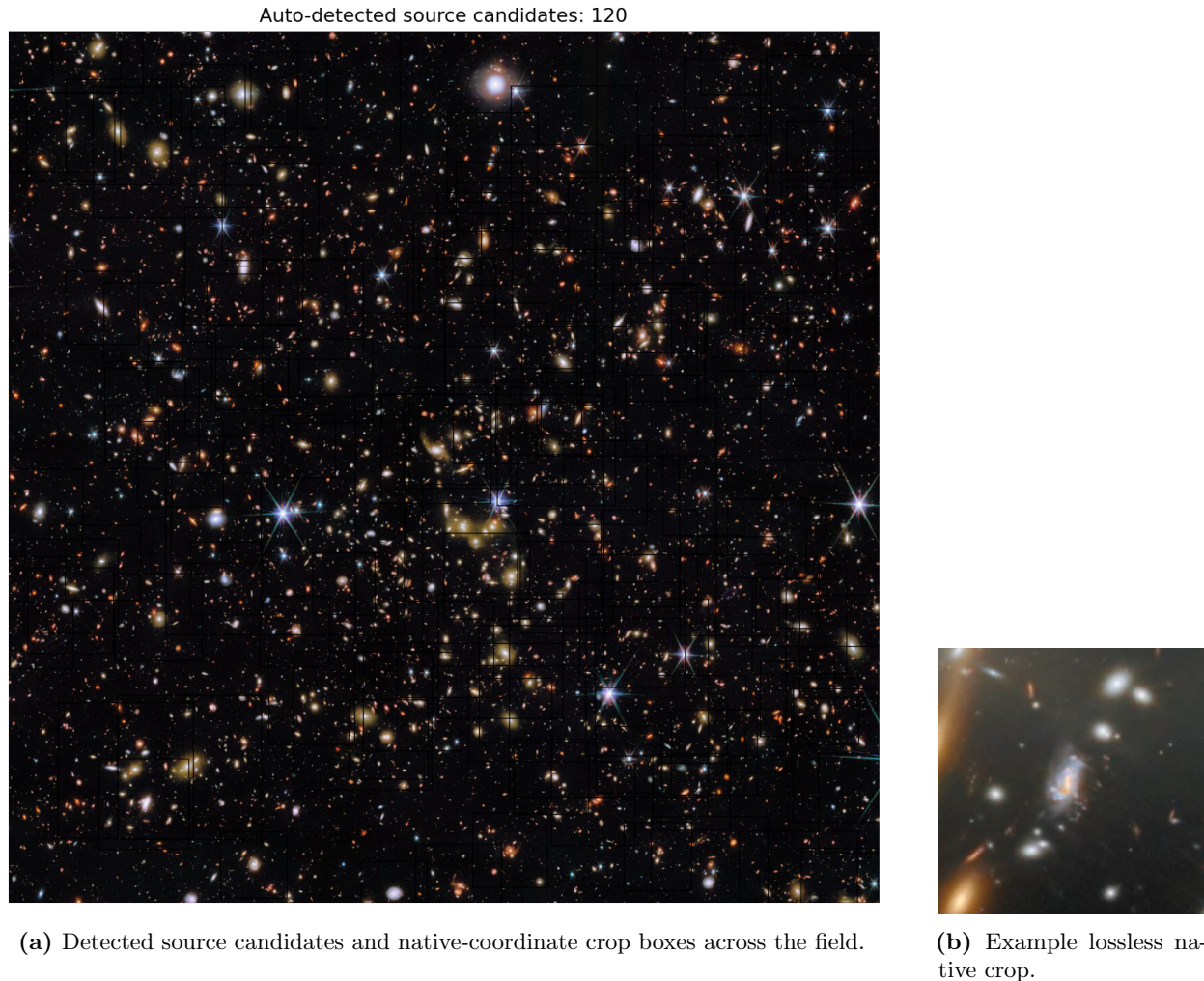


Figure 3: Batch-processing stage. Detection may use a resized working copy for speed, but the boxes are mapped back to the original image before crops are saved. The black rectangles therefore represent native-coordinate extraction regions, not resized thumbnails.

For speed, the batch program first performs a cheaper morphology estimate. The expensive 30-trial robust handedness analysis is normally run only when the quick morphology branch identifies `Spiral Galaxy` as the leading candidate. The program writes a CSV catalog containing crop coordinates, morphology scores, b/a , r_{50} , r_{90} , SNR proxy, concentration, asymmetry, clumpiness, bright-component count, handedness diagnostics and quality flags.

The pilot configuration intentionally capped processing at the 120 largest detected candidates. This cap makes the run manageable for development, but it also means the sample is not a complete flux-limited or volume-limited galaxy sample.

6 Pilot results

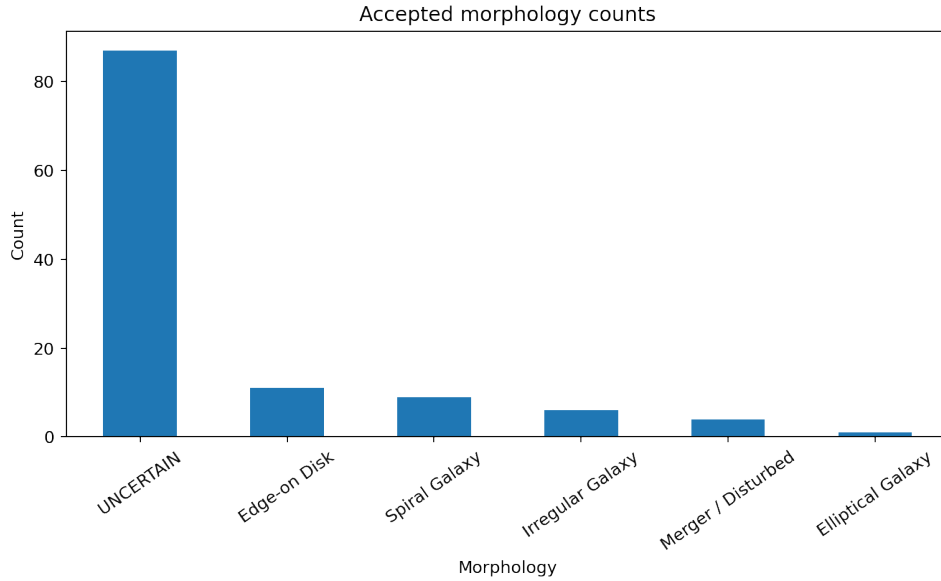
6.1 Morphology catalog

The batch run produced 120 candidate rows. Two rows ended in processing errors, leaving 118 objects in the morphology summary. The conservative compact-source flag did not label any object as star-like in this particular run; this should not be interpreted as proof of zero stellar or artifact contamination, because the flag is only a simple size–concentration heuristic.

Table 2 gives the final morphology labels. The dominant result is `UNCERTAIN`: 87 of 118 objects, or 73.7%. This is not surprising for small, blended and strongly processed deep-field sources, and it demonstrates the intended effect of the score and margin vetoes.

Table 2: Final morphology labels for the 118 objects included in the pilot statistics.

Final label	Count	Fraction
UNCERTAIN	87	73.7%
Edge-on Disk	11	9.3%
Spiral Galaxy	9	7.6%
Irregular Galaxy	6	5.1%
Merger / Disturbed	4	3.4%
Elliptical Galaxy	1	0.8%

**Figure 4:** Final morphology counts from the 118-object pilot. Most sources remain uncertain after the conservative top-score and score-margin criteria are applied.

The best-candidate labels before the uncertainty veto were different: 65 merger/disturbed, 19 edge-on, 15 irregular, 10 elliptical and 9 spiral candidates. The large difference between “best candidate” and accepted final class shows why reporting only the maximum score would be misleading. Many objects have two or more plausible explanations under the present heuristic model.

6.2 Spiral handedness

Sixteen objects entered the robust handedness stage. None passed every threshold in Table 1; all 16 therefore remained **UNCERTAIN** in winding, with seven diagnostic CW signs and nine diagnostic CCW signs. Restricting attention to the nine objects whose final morphology was accepted as **Spiral Galaxy**, the diagnostic split was four CW and five CCW, again with zero strict acceptances.

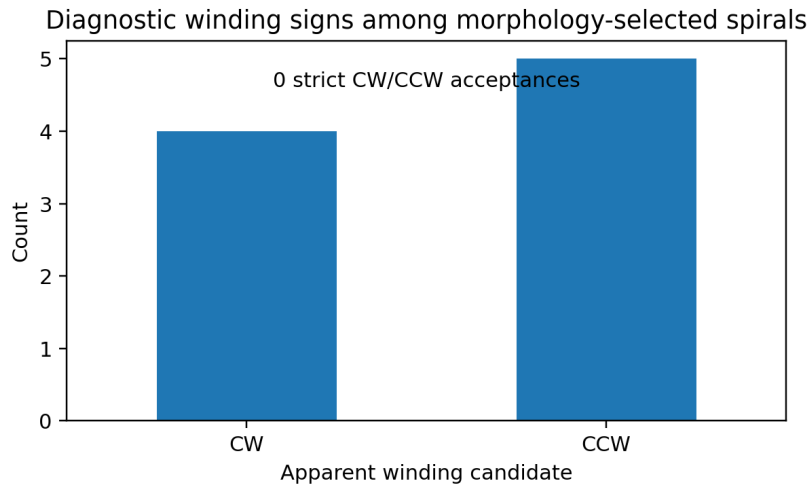


Figure 5: Diagnostic winding signs among the nine objects accepted morphologically as spiral galaxies. Four favor CW and five favor CCW, but none satisfies every strict handedness criterion. The bars therefore show tentative signs, not accepted directions.

The failure diagnostics are informative. Among the nine morphology-selected spirals, all nine had an explained-structure fraction below 0.20; eight had $|H| < 0.35$; five had parameter stability below 0.90; five had spiral/straight response below 1.30; and two had coherence below 0.45. None failed the noise-stability threshold and none was rejected for $b/a < 0.45$. The median spiral/straight response was 1.29, close to the adopted cutoff, but the median explained fraction was only 0.051. In other words, the program often found a signed pattern that was repeatable under added noise, yet that pattern accounted for too little of the total non-axisymmetric structure to justify a strong chirality claim.

This is a useful negative result. A less conservative program could easily have converted these weak positive or negative H values into a visually convincing CW/CCW histogram. The present pipeline instead reports that the data and method do not yet support such a conclusion.

7 Discussion

7.1 What changed during development?

The largest conceptual change was the distinction between *enhancement* and *measurement*. The first experiments were dominated by the question “how can I make the arms visible?” The final pipeline asks a different sequence of questions: is a source detected; is it resolved; what is its projected shape; how asymmetric and clumpy is its light; is a spiral template better than a straight one; does the sign survive changes in center, annulus and smoothing; and does mirroring reverse the answer as it should?

This change also made the code easier to debug. Every accepted or rejected result can be traced to numerical diagnostics rather than only to a processed image. For example, a visually spiral-looking crop can still fail because r_{90} is too small, because the straight-template response is comparable to the spiral response, or because the sign changes when the center moves by less than a pixel.

The morphology branch and the handedness branch also serve different purposes. Morphology asks for the best broad structural description, whereas handedness is a more demanding signed measurement that should only be trusted when spiral evidence is coherent. Combining them in a single pipeline does not mean that every object must receive both answers.

7.2 Current limitations

The first limitation is the input data. Public JWST color composites are scientifically informed visual products, but they undergo stretching, filter combination, alignment and sometimes resampling for presentation [5]. Such images are valuable for algorithm development, yet a quantitative morphology study should eventually use calibrated science products, preferably common-grid mosaics with documented pixel scale, PSF and uncertainty information [6].

Second, the detector is a source detector rather than a full astronomical deblender. Close galaxies, gravitational arcs, stellar diffraction patterns and bright foreground sources can merge into a connected component. The present “possible star-like” flag is too simple to solve star–galaxy separation. A future catalog should include explicit PSF-based compact-source rejection and segmentation/deblending before morphology is measured.

Third, the morphology scores are hand-designed. Their strength is interpretability, but their numerical weights and decision thresholds were not fitted on a labeled dataset. Validation against a trusted reference sample is therefore necessary before quoting accuracy, completeness or contamination. Galaxy Zoo-type labels would be a natural comparison set, while maintaining care about human handedness biases [4].

Fourth, the pilot is selection-limited. It processed the 120 largest source candidates produced by one detector configuration, not every astronomical object in the field. The resulting fractions in Table 2 describe the program’s pilot output, not the underlying galaxy population.

Finally, apparent winding should not be conflated with physical spin direction. Dust geometry, projection and the near/far side of a disk are additional physical questions. The present code measures only image-plane chirality.

8 Conclusion

This project began with Gaussian subtraction, sharpening and edge maps applied to individual low-resolution galaxy crops. Those experiments were valuable because their artifacts made the limitations of purely visual enhancement obvious. The code then evolved into a quantitative, interpretable pipeline: robust sky estimation, connected-source isolation, size and shape measurements, CAS-inspired structural proxies, five-class morphology scores, a log-polar Fourier spiral test, parameter perturbations, noise stress tests and mirror validation. The final extension automatically detects many sources in a full field, crops them from the native-resolution original and writes a reproducible catalog.

The pilot results are conservative. Of 118 analyzed objects, 87 remained morphologically uncertain, while 31 received one of five structural labels. Nine were accepted as spiral morphologies, but no object passed all strict CW/CCW requirements. This should be viewed as a strength of the present baseline rather than a failure: the pipeline distinguishes a diagnostic preference from an accepted measurement.

The next stage is clear. The same code should be run on calibrated JWST FITS mosaics, coupled to better deblending and compact-source rejection, and validated on externally labeled galaxies. Once that baseline is quantified, a learned CNN or vision-transformer classifier can be added as a second branch and compared against the interpretable measurements rather than replacing them. The long-term goal is therefore not simply to “enhance” distant galaxies, but to build a system whose morphology and handedness decisions can be audited, stress-tested and scientifically calibrated.

A Pipeline summary and reproducibility

The complete batch workflow used in the pilot can be summarized as:

```
full field
-> robust luminance + local-background residual
-> connected-component source detection
```

```

-> map boxes to native coordinates
-> lossless PNG crops from original image
-> source detection / resolution checks
-> b/a, r20/r50/r80/r90, SNR, C, A, clumpiness
-> quick five-class morphology score
-> robust log-polar Fourier stage for spiral candidates
-> 30 parameter trials + 8 noise trials + mirror test
-> final morphology and CW/CCW/UNCERTAIN labels
-> CSV catalog + count tables + diagnostic figures

```

The implementation uses Python with NumPy, SciPy, Pillow, Matplotlib and pandas. The early denoising experiment additionally used `skimage.restoration.denoise_bilateral`. In the pilot configuration, source detection used a residual threshold of 2.8 MAD-equivalent noise units, a crop scale of 3.2, native crop sides between 48 and 500 pixels, and a cap of 120 largest candidates. The robust spiral stage used eight added-noise trials in the batch run. Crops were saved as PNG directly from the original image without resizing.

B Reference values from the nine accepted spiral morphologies

Table 3 reports the median diagnostics for the nine objects whose final morphology was **Spiral Galaxy**. These medians summarize why morphology could be accepted while handedness could not.

Table 3: Median robust diagnostics for the nine morphology-selected spiral galaxies.

Diagnostic	Median
Signed winding score H	−0.126
Parameter stability	0.867
Noise stability	1.000
Phase coherence	0.529
Explained residual structure	0.051
Spiral/straight response	1.289
Projected axis ratio b/a	0.660

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